

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2.	(a)			Gives charge [electrical potential] energy <i>Accept</i> pumps charge around circuit / does work on charge. <i>Accept</i> electrons instead of charge	1			1		
	(b)	(i)		Total emf = 3×1.60 or 4.80 [V] (1) Total resistance = $(1.20 + (3 \times 0.10))$ or 1.50 [Ω] (1) Current $[= \frac{V}{R}] = 3.2$ [A] (1) N.B. If workings in reverse award a maximum of 2 marks		3		3	1	
		(ii)	I	12.3 [W] ecf (10.8 W if 3 A used) e.g. using $P = I^2 R = 3.2^2 \times 1.2 = 12.3$ [W]		1		1	1	
			II	15.4 [W] ecf (14.4 W if 3 A used) e.g. using $P = IE = 3.2 \times 4.8 = 15.4$ [W]		1		1	1	
		(iii)		Thermal or internal. <i>Accept</i> 'heat', 'dissipated' (1) in internal resistance (1)	2			2		
	(c)			3.15 A or 3.2 A [<i>Accept</i> : even on short-circuit, extra cell gives only 3.0 A] (1) <u>No.</u> Current less than [<i>accept</i> same as] before ecf on (b)(i) (1)			2	2	1	
				Question 2 total	3	5	2	10	4	0